

Speculative Self-Distillation enables Efficient Knowledge Internalization

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PAPER

TL;DR

Problem. Self-distillation puts a document into a model's weights, but **on-policy** training wastes most of its tokens.

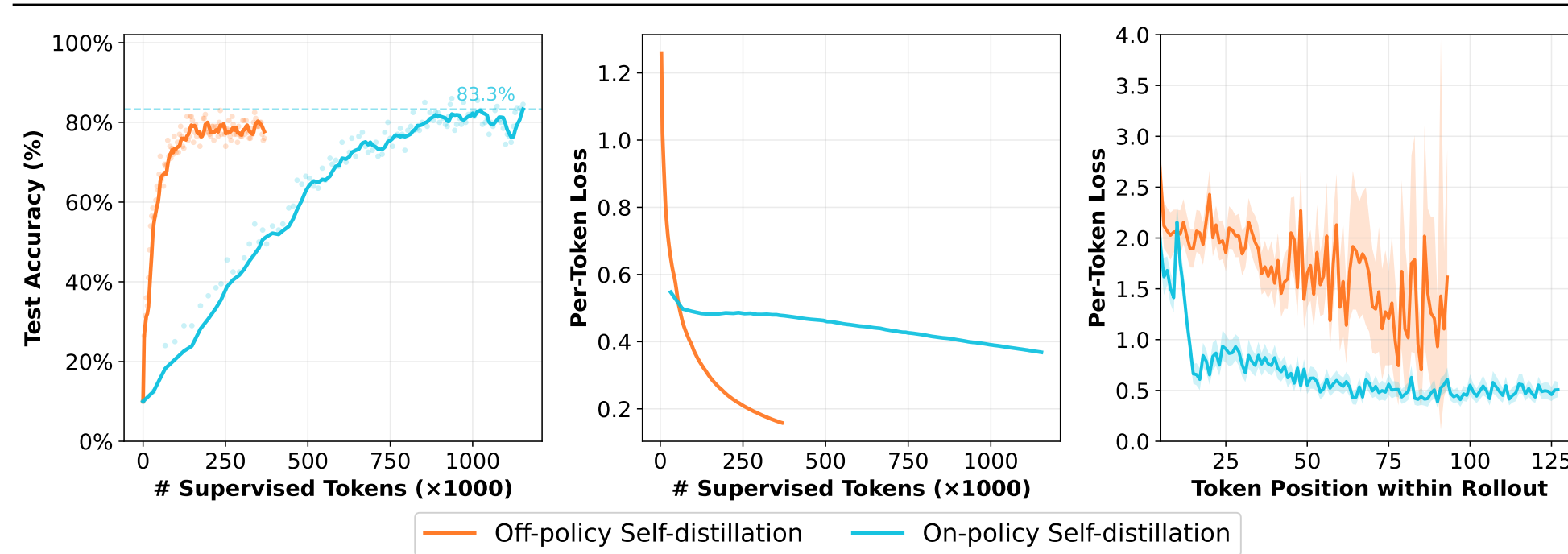
Method. **SSD**: the student writes the rollout; the teacher takes over only where the two disagree.

Result. **on-policy** accuracy with **45% fewer supervised tokens**, and less forgetting.

Self-Distillation

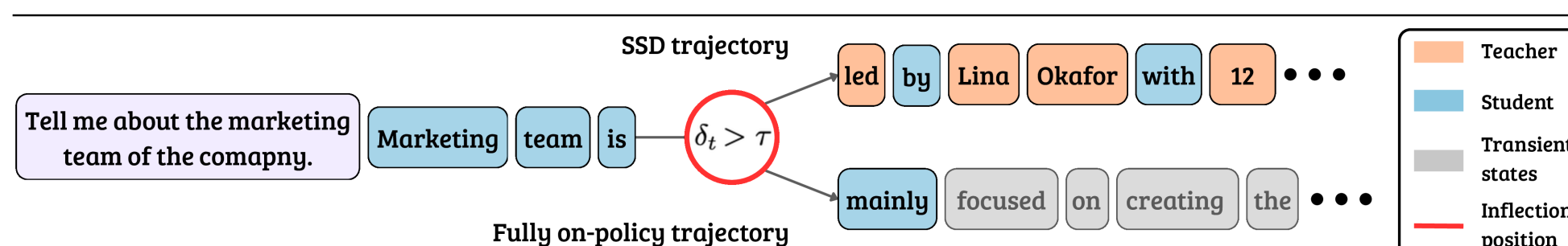
- **Teacher**: the model with the document in context.
Student: the same model without it.
- The student is trained to match the teacher token by token.
- **off-policy**: teacher writes the rollout. **on-policy** (SDFT): student writes it.

Why On-Policy Is Slow



- **off-policy** learns $\sim 5\times$ faster but plateaus lower.
- Along **on-policy** rollouts the per-token signal collapses after a few tokens.

Inflection Tokens

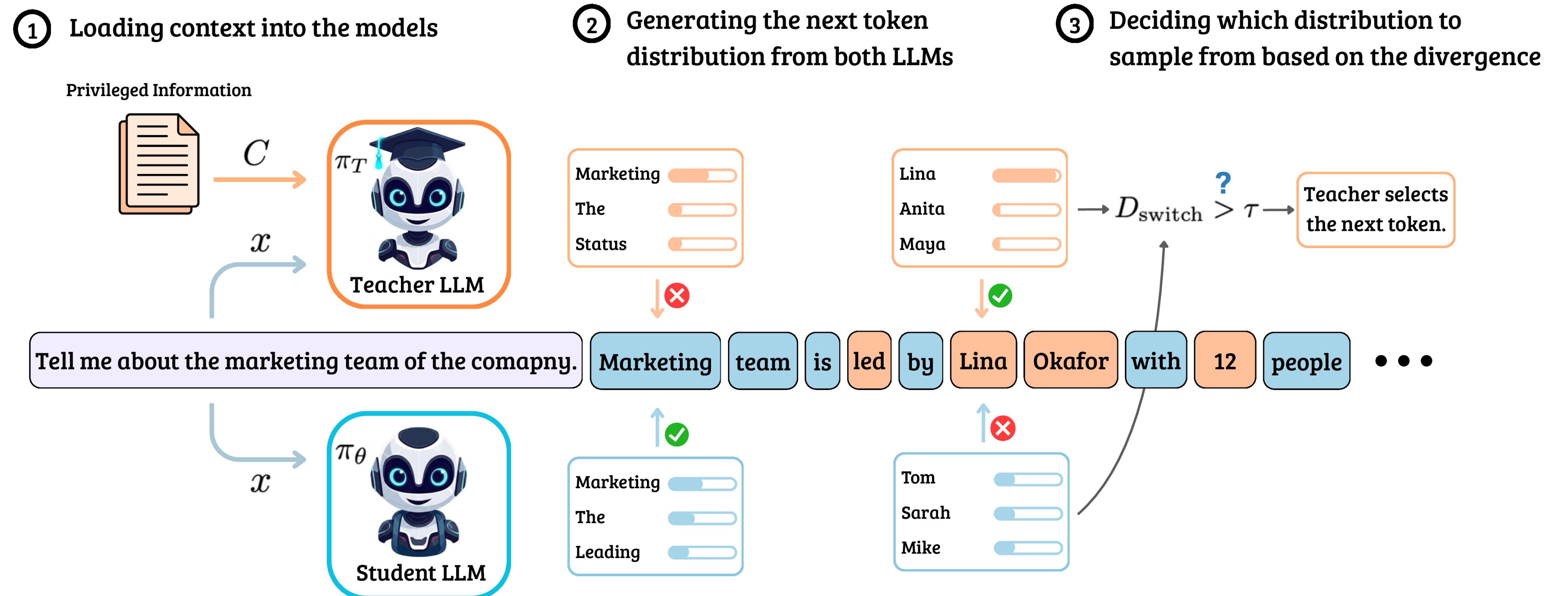


- Where the document changes the next token ($\delta_t > \tau$), the teacher writes.
- **on-policy** keeps the student's token and drifts into **transient states**.

Setup

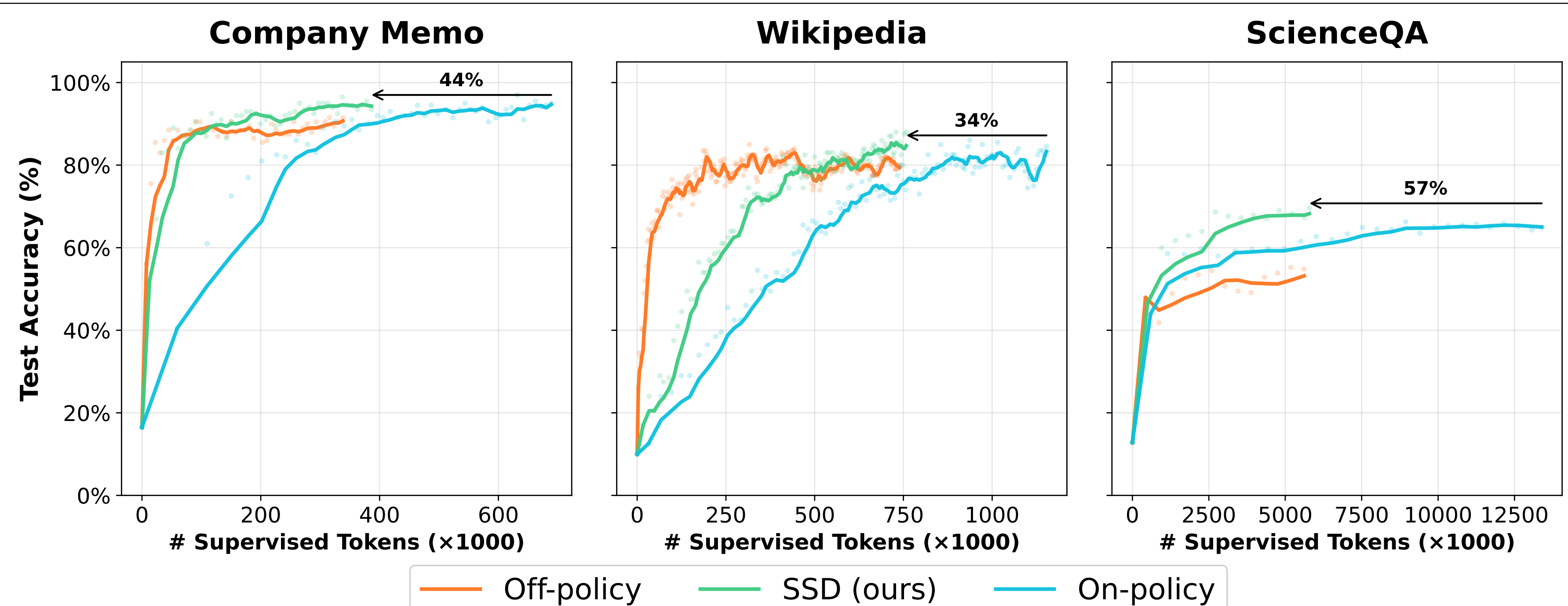
- **Model**: Qwen3-4B-Instruct as student and (frozen) teacher.
- **Tasks**: Company Memo, Wikipedia, Science Q&A (closed-book QA).
- **Cost**: supervised tokens, i.e. positions the loss is computed on.

Speculative Self-Distillation (SSD)



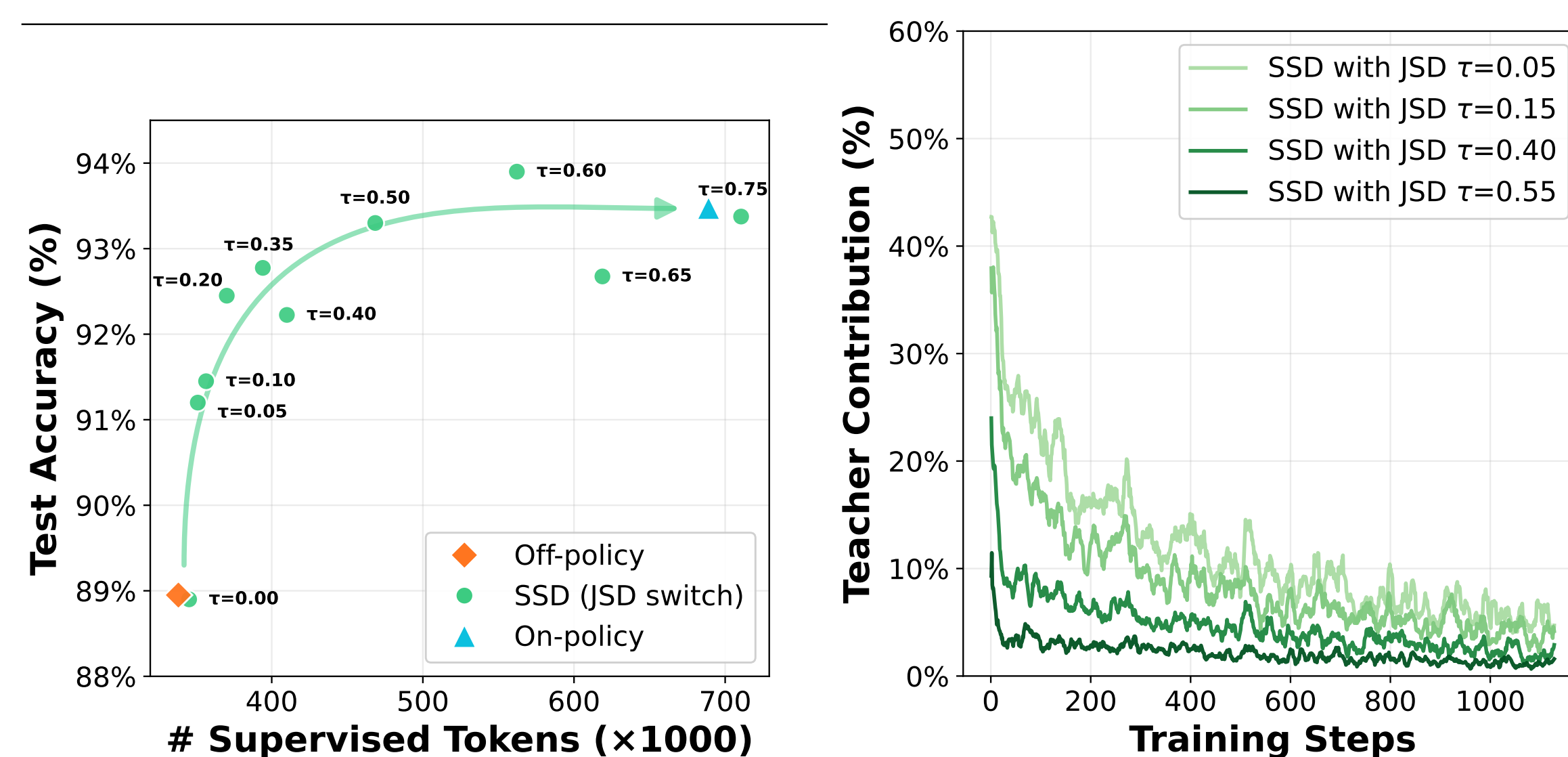
- Student generates by default.
- If $D_{\text{switch}}(\pi_\theta \parallel \pi_T) > \tau$, the teacher picks the token.
- Every position is trained with the same loss.
- $\tau \rightarrow 0$ is **off-policy**, $\tau \rightarrow \infty$ is **on-policy**.

SSD Reaches On-Policy Accuracy with 34–57% Fewer Tokens



Qwen3-4B-Instruct as student and teacher, three closed-book QA tasks.

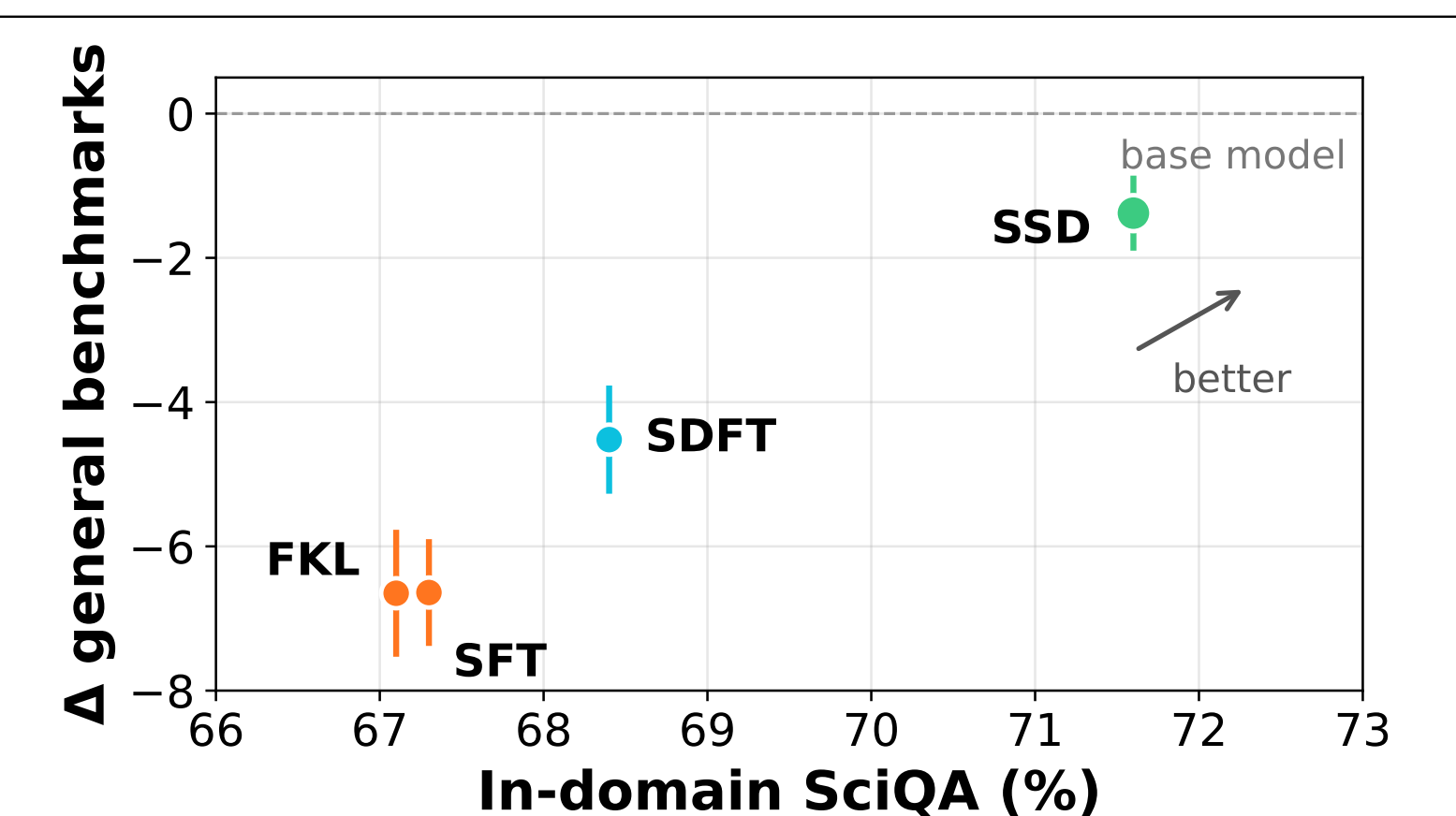
One Knob, and a Built-in Curriculum



$\tau \in [0.2, 0.5]$ sits on the Pareto frontier.

The teacher's share fades with no schedule.

Less Forgetting



Qwen2.5-7B, SciKnowEval chemistry: best in-domain accuracy, smallest drop on six general benchmarks.